



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES

Digital

Part A: Content Design

Course Title	Introduction to Data Science
Course No(s)	DSE ZG532
Credit Units	5
Credit model	3+1+1
Instructor	S PRABAKERAN
Content Authors	Prof. A Vijayalakshmi
Version	2.0
Date	11-11-2025

Course Descriptions

Data Analytics, Data and Data Models, Data wrangling, Feature Engineering, Classification and Prediction, Association Analysis, Clustering, Anomaly Detection, exploratory/explanatory data analysis with visual storytelling, Ethics for Data Science.

Course Objectives

No	Course Objective
C01	Gain basic understanding of the role of Data Science in various scenarios in the real-world of business, industry and government.
C02	Understand various roles and stages in a Data Science Project and ethical issues to be considered.
C03	Explore the processes, tools and technologies for collection and analysis of structured and unstructured data.
C04	Appreciate the importance of techniques like data visualization, storytelling with data for the effective presentations of the outcomes with the stakeholders
C05	Understand techniques of preparing real-world data for data analytics.

C06	Implement data analytic techniques for discovering interesting patterns from data.
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Text Book(s)

T1	Introduction to Data Mining, by Tan, Steinbach and Vipin Kumar
T2	Introducing Data Science by Cielen, Meysman and Ali
T3	Storytelling with Data, A data visualization guide for business professionals, by Cole Nussbaumer Knaflitz; Wiley
T4	Data Mining: Concepts and Techniques, Third Edition by Jiawei Han and Micheline Kamber Morgan Kaufmann Publishers, 2006

Reference Book(s) & other resources

R1	The Art of Data Science by Roger D Peng and Elizabeth Matsui
R2	Ethics and Data Science by DJ Patil, Hilary Mason, Mike Loukides
R3	Python Data Science Handbook: Essential tools for working with data by Jake VanderPlas
R4	KDD, SEMMA and CRISP-DM: A Parallel Overview , Ana Azevedo and M.F. Santos , IADS-DM, 2008
R5	A Beginner's Guide to Data & Analytics by Harvard Business School
R6	On Being a Data Skeptic Publisher(s): O'Reilly Media, Inc. ISBN: 9781449374310

Learning Objectives: Students will be able to

L01	Learn the data science lifecycle, including data collection, processing, analysis, and visualization.
L02	Understand fundamental machine learning models(classification and clustering)
L03	Identify patterns, trends, and anomalies in datasets.

Content Structure

- 1 Introduction to Data Science (2 hrs)
 - 1.1 Fundamentals of Data Science
 - 1.2 Real World applications
 - 1.3 Data Science Challenges
 - 1.4 Data Science Teams
 - 1.4.1 Defining Data Team
 - 1.4.2 Roles in a Data Science Team
 - 1.4.3 Roles and responsibilities of a Data Scientist, Data Engineers
 - 1.5 Software Engineering for Data Science
 - 1.5.1 DataOps
 - 1.5.2 MLOps (intro)
2. Data Analytics (2hrs)
 - 2.1. Defining Analytics
 - 2.2. Types of data analytics
 - 2.2.1. Descriptive, Diagnostic, Predictive, Prescriptive
 - 2.3. Data Science Process
 - 2.3.1. CRISP-DM Methodology
 - 2.3.2. SEMMA
 - 2.3.3. BIG DATA LIFE CYCLE
 - 2.3.4. SMAM
3. Data and Data Models (2 hrs)
 - 3.1. Types of Data and Datasets
 - 3.2. Data Quality and Issues: An overview
 - 3.3. Data Models
 - 3.3.1. General Framework of Formal modeling
 - 3.3.2. Association Analyses
 - 3.3.3. Prediction Analyses
 - 3.4. Data Pipelines and patterns;
 - 3.5. Data Pipeline Stages
 - 3.5.1 Data extraction
 - 3.5.2 Data ingestion
 - 3.5.3 Data cleaning
 - 3.5.4 Data exploration
 - 3.5.5 Data wrangling
 - 3.5.6 Data versioning
 - 3.5.7 Data transformation
 - 3.5.8 Feature management
 - 3.6 Modern Data Infrastructure
 - 3.6.1 Diverse data sources
 - 3.6.2 Cloud data warehouses and lakes
4. Data wrangling (4 hrs)
 - 4.1 Data cleaning
 - 4.2 Data Aggregation, Sampling,

- 4.3 Statistical descriptions of data
- 4.4 Measuring data similarity & dissimilarity
- 4.5. Handling Numeric Data
 - 4.5.1 Discretization, Binarization
 - 4.5.2 Normalization
 - 4.5.3 Data Smoothing
- 4.6 Dealing with textual Data
- 4.7 Dealing with Images, audio and video data
- 4.8. Managing Categorical Attributes
 - 4.8.1 Transforming Categorical to Numerical Values
 - 4.8.2 Encoding techniques
- 4.9 Overview of visualization techniques for Data Exploratory analysis
- 5. Feature Engineering (4 hrs)
 - 5.1 Feature Extraction
 - 5.2 Feature Construction
 - 5.3 Feature Subset selection
 - 5.3.1 Filter methods
 - 5.3.2 Wrapper methods
 - 5.3.3 Embedded methods
 - 5.4 Feature Learning
 - 5.5 Feature reduction (Dimensionality Reduction)
 - 5.6 Case Study involving FE tasks
 - 5.7 Feature Engineering techniques for text, images, audio, video
- 6. Classification and Prediction (4 hrs)
 - 6.1. Concepts of classification and prediction
 - 6.2. Decision trees for classification - ID3 algorithm using entropy and Gini Index
 - 6.3. Evaluation of classification algorithms
- 7. Association Analysis (4 hrs)
 - 7.1. Association analysis concepts
 - 7.2. Apriori Algorithm for frequent itemsets
 - 7.3 FP Growth for frequent itemsets
 - 7.4. Mining association rules
- 8. Clustering (4 hrs)
 - 8.1. Cluster analysis concepts.
 - 8.2. Partitioning methods – k-Means algorithm
 - 8.3. Hierarchical methods for cluster analysis
 - 8.4. Density based methods for cluster analysis - DBSCAN
 - 8.5. Evaluation of clustering algorithms
- 9. Anomaly Detection (2 hr)
 - 9.1. Concepts of Outliers
 - 9.2. Statistical approaches
 - 9.3. Proximity and Density based outlier detection
- 10. Storytelling with Data (2 hr)

- 10.1. The final deliverable
- 10.2. The Narrative - report / presentation structure
- 10.3. Building narrative with Data
- 10.4. Effective storytelling

11. Ethics for Data Science (2 hr)

- 11.1. Bias and Fairness in Data
- 11.2 Being a data skeptic – examples of misuse of Data
- 11.3 Five C's
- 11.4 Ethical guidelines for Data Scientist
- 11.5 Ethics of data scraping and storage
- 11.6 Case Study: IBM AI Fairness 360

Part B: Learning Plan

Academic Term	
Course Title	Introduction to Data Science
Course No	
Lead Instructor	

Session No.	Topic Title	Resource Reference
1	Introduction to Data Science (2 hrs) ● Fundamentals of Data Science ● Real World applications ● Data Science Challenges ● Data Science Teams ◦ Defining Data Team ◦ Roles in a Data Science Team ◦ Roles and responsibilities of a Data Scientist, Data Engineers ● Software Engineering for Data Science ◦ DataOps ◦ MLOps (intro)	T3 – Ch 1 T4 – Ch1 T1 – Ch1 Class Room Discussion Class Notes Additional Reading (AR) material provided LMS
2	Data Analytics (2hrs) ● Defining Analytics ● Types of data analytics ◦ Descriptive, Diagnostic, Predictive, Prescriptive ● Data Science Process ◦ CRISP-DM Methodology ◦ SEMMA ◦ BIG DATA LIFE CYCLE	R5 R4

	◦ SMAM	
3	Data and Data Models (2 hrs) • Types of Data and Datasets • Data Quality and Issues: An overview • Data Models ◦ General Framework of Formal modeling ◦ Association Analyses ◦ Prediction Analyses • Data Pipelines and patterns • Data Pipeline Stages • Data extraction • Data ingestion • Data cleaning • Data exploration • Data wrangling • Data versioning • Data transformation • Modern Data Infrastructure ◦ Diverse data sources ◦ Cloud data warehouses and lakes	T1 – Ch 2.1, 2.2 R1 – Ch 2, Ch 7 Class room discussions
4	Data wrangling (2 hrs) • Data cleaning • Data Aggregation, Sampling, • Statistical descriptions of data • Measuring data similarity & dissimilarity • Handling Numeric Data ◦ Discretization, Binarization ◦ Normalization ◦ Data Smoothing	T1 – Ch2.3, 2.4 T4 – Ch4

5	Data wrangling (2 hrs) • Dealing with textual Data • Dealing with Images, audio and video data • Managing Categorical Attributes ◦ Transforming Categorical to Numerical Values ◦ Encoding techniques • Overview of visualization techniques for Data Exploratory analysis	Class room discussions T1 – Ch3.1
6	Feature Engineering (2 hrs) • Feature Extraction • Feature Construction • Feature Subset selection ◦ Filter methods ◦ Wrapper methods ◦ Embedded methods • Feature reduction (Dimensionality Reduction) (PS: PCA discussed in depth in MFDS course of DSE programme.)	T1 – Ch2 Class room discussions T1 – Appendix B.1
7	Feature Engineering (2 hrs) • Case Study involving FE tasks • Feature Engineering techniques for text, images, audio, video	Class room discussions T4 – Ch10.4
8	Classification and Prediction (2 hrs) • Concepts of classification and prediction • Evaluation of classification algorithms	T4 – Ch6.1, 6.2, 6.3 T4 – 6.12, 6.13, 6.15
9	Classification and Prediction (2 hrs) • Decision trees for classification - ID3 algorithm using entropy and Gini Index, Occam's razor (Mutual Information and Gini Index are used as Feature subset selection)	T4 – Ch6.1, 6.2, 6.3

	techniques.)	
10	Association Analysis (2 hrs) • Association analysis concepts • Apriori Algorithm for frequent itemsets	T1 – Ch6 T4 – Ch5
11	Association Analysis (2 hrs) • FP Growth for frequent itemsets • Mining association rules	T1 – Ch6 T4 – Ch5
12	Clustering (2 hrs) • Cluster analysis concepts. • Partitioning methods – k-Means algorithm • Hierarchical methods for cluster analysis	T1 – Ch8 T4 – Ch7
13	Clustering (2 hrs) • Density based methods for cluster analysis – DBSCAN • Evaluation of clustering algorithms	T1 – Ch8 T4 – Ch7 T1 – Ch8.5
14	Anomaly Detection (2 hrs) • Concepts of Outliers • Statistical approaches • Proximity and Density based outlier detection	T1 – Ch10 T4 – Ch7.11
15	Storytelling with Data (2 hrs) • The final deliverable • The Narrative - report / presentation structure • Building narrative with Data • Effective storytelling	T3 – Ch10
16	Ethics for Data Science (2 hrs) • Bias and Fairness	R6 T4 – Ch11.4

	◦ Types of Bias ◦ Identifying Bias ◦ Evaluating Bias ● Being a data skeptic – examples of misuse of Data ● Five C's ● Ethical guidelines for Data Scientist ● Ethics of data scraping and storage ● Case Study: IBM AI Fairness 360 (PS: Ethics for Data is the focus.)	R2 – Ch1, Ch3
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Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc...) number of lab exercises needed, etc.

1. Lab work: 9
2. Project work: 0
3. Case Study: 4 Webinars
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of fundamental NLP algorithms using state of art tools

Scope of Experiential Learning Component:

Programming language - Python

Tools and libraries: NLTK, Scipy ,Matplotlib ,Seaborn etc

Lab Infrastructure:

Online/ Open source/Google Colab

Detailed Plan for Lab work

Lab No.	Lab Objective	Lab Sheet Access URL	Session Reference
1	Introduction to Python, Numpy, Scipy, Python Pandas,		2
2	Data ingestion and extraction, data aggregation techniques		3
3	Exploration and Visualizing using Matplotlib, Seaborn		4

4	Data pre-processing in Python - Discretization, Binarization, Normalization, Data Smoothing, Managing Categorical Attributes		5
5	Feature Engineering using Filter methods, wrapper methods, PCA		7
6	Data pre-processing and Feature Engineering techniques for text, images, audio, video		8
7	Decision trees for classification using Scikit learn		9
8	Association Analysis using Scikit learn		11
9	Clustering analysis by kmeans, hierarchical methods, DBScan using Scikit learn		13

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

No	Name	Type	Duration	Weight	Day, Date, Session, Time
EC-1(a)	Quizzes	Online		10%	August 10-20, 2026
EC-1(b)	Assignments	Take Home		20%	August 27-7 Sept 2026
EC-2	Mid-Semester Test	Closed Book		30%	20/09/2026 (FN)
EC-3	Comprehensive Exam	Open Book		40%	06/12/2026 (FN)

Note:

Syllabus for Mid-Semester Test (Closed Book): Topics in Session Nos. 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics (Session Nos. 1 to 16)

Important links and information:

Elearn portal: <https://elearn.bits-pilani.ac.in> or Canvas

Students are expected to visit the Elearn portal on a regular basis and stay up to date with the latest announcements and deadlines.

Contact sessions: Students should attend the online lectures as per the schedule provided on the Elearn portal.

Evaluation Guidelines:

- 1 EC-1 consists of two Quizzes. Students will attempt them through the course pages on the Elearn portal. Announcements will be made on the portal, in a timely manner.
- 2 EC-2 consists of either one or two Assignments. Students will attempt them through the course pages on the Elearn portal. Announcements will be made on the portal, in a timely manner.
- 3 For Closed Book tests: No books or reference material of any kind will be permitted.
- 4 For Open Book exams: Use of books and any printed / written reference material (filed or bound) is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
- 5 If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam which will be made available on the Elearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course hand-out, attend the online lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the hand-out.